

From the labs

With artificial wombs a possibility, are we ready to play with the power of life?

**Microsoft Chromebook**

Reportedly, new Windows 10 Cloud OS to go against Google's Chromebooks.
<http://digit.in/Win10cloud>

FAREWELL CASSINI

Chronicling the journey of the Cassini-Huygens mission across the outer solar system.

Abhijit Dey
abhijit@digit.in

NASA's Cassini mission is nearing its end. The decade-old spacecraft is preparing for its final dive into Saturn. The unmanned spacecraft will carry out its ultimate mission of studying more about Saturn while analysing the gap between the planet's upper atmosphere and the innermost ring. Plunging the spacecraft to its death was always being considered and as the spacecraft has almost depleted its fuel resources, the motivation to carry out the final study makes more sense now. Cassini will begin the final spiral towards the ringed-planet on April 22. By September 15, or in about four months from now, there won't be much of the spacecraft left, as it will get shredded by Saturn's harsh atmosphere.

Humble beginnings

Launched back on October 15, 1997, the Cassini-Huygens mission was primarily deployed to study Saturn and its satellites. It consisted of two modules including Cassini, the Saturn orbiter, and Huygens, the moon lander for Saturn's biggest moon Titan. Both the spacecraft entered Saturn's orbit on July 1, 2004, and after a few months on December 25,

2004, they separated and Huygens landed on Titan on January 14, 2005.

The Cassini-Huygens mission was a joint space mission between NASA, the European Space Agency, and the Italian Space Agency. The mission, which is still ongoing, has a lot of achievements to its name that we will be talking about in detail. Out of the 62 moons of Saturn,

two particular moons have drawn the interest of astronomers and scientists – Titan and Enceladus.

Life on Titan in the future

As already mentioned, the primary mission of the European Space Agency's Huygens probe was to study Saturn up close and land on Titan. This landing marked the first ever successful mission of landing on a moon in the outer solar system. On touchdown, the surface of Saturn's biggest moon was found to be an icy terrain with surface temperatures at a freezing minus 179 degrees Celsius. Titan consists of rivers, seas and lakes filled with liquid methane and ethane. Just like Earth's water cycle, the moon also displays a similar weather pattern of rainfall. The same process of evaporation and condensation occurs. Among the methane seas on the moon, the second largest sea Ligeia Mare was determined to have a depth of 170 metres. This was the first time we were able to calculate the depth of a liquid body on



Cassini in the space simulator

Credit: NASA / Jet Propulsion Laboratory